

- **a) Encrypting data before sending: Presentation Layer (Layer 6)**
Reason: The Presentation Layer prepares data before transmission. It handles **encryption, decryption, compression, and data format conversion**.
Example: TLS/SSL encryption used when accessing a secure HTTPS website.
- **b) Routing packets across networks: Network Layer (Layer 3)**
Reason: The Network Layer finds the **best path** for packets to travel between different networks. It uses **IP addresses** for logical addressing.
Example: A **router** forwards packets from your home network to another network on the internet.
- **c) Converting data to electrical signals: Physical Layer (Layer 1)**
Reason: The Physical Layer transmits raw **bits (0s and 1s)** through physical media by converting them into electrical, optical, or radio signals.
Example: An Ethernet cable carries electrical signals.
- **d) Opening a web page in your browser: Application Layer (Layer 7)**
Reason: The Application Layer provides network services directly to user applications such as web browsers.
Example: Your browser uses **HTTP or HTTPS** to request a web page.
- **e) Detecting and correcting errors in frames: Data Link Layer (Layer 2)**
Reason: The Data Link Layer works with **frames** and handles local network delivery, MAC addressing, and frame error detection.
Example: Ethernet can use **FCS/CRC** to detect corrupted frames.